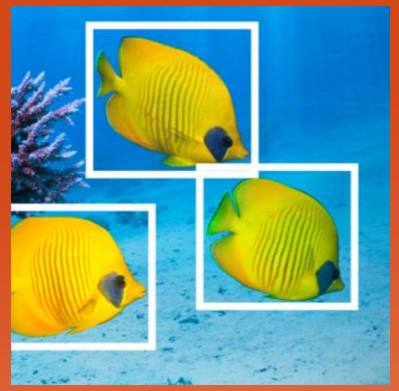


Cutting Labeling Costs in an Imbalanced Seafood Dataset: An Active Learning Approach with ResNet-18

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Final Project for AI 221

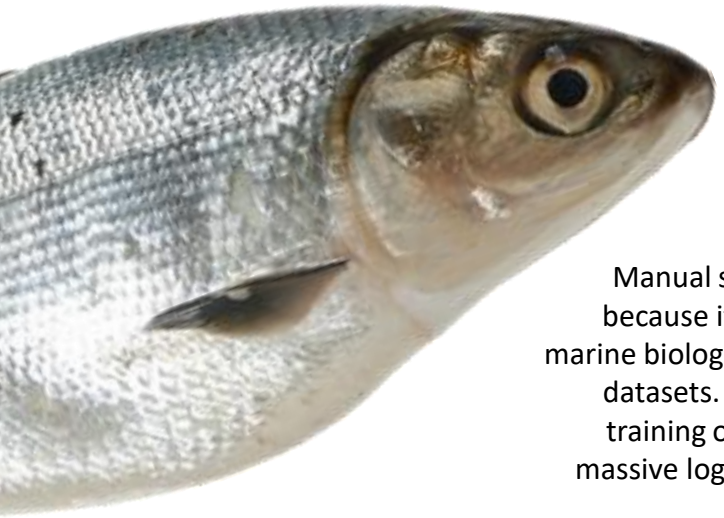
Introduction

Automatic seafood classification is interesting because it replaces the slow, subjective, and error-prone process of manual labelling.

- By automating classification using AI systems, agencies can better monitor biodiversity, stop illegal fishing, and streamline industrial sorting.

Problem: Costly Labeled Data

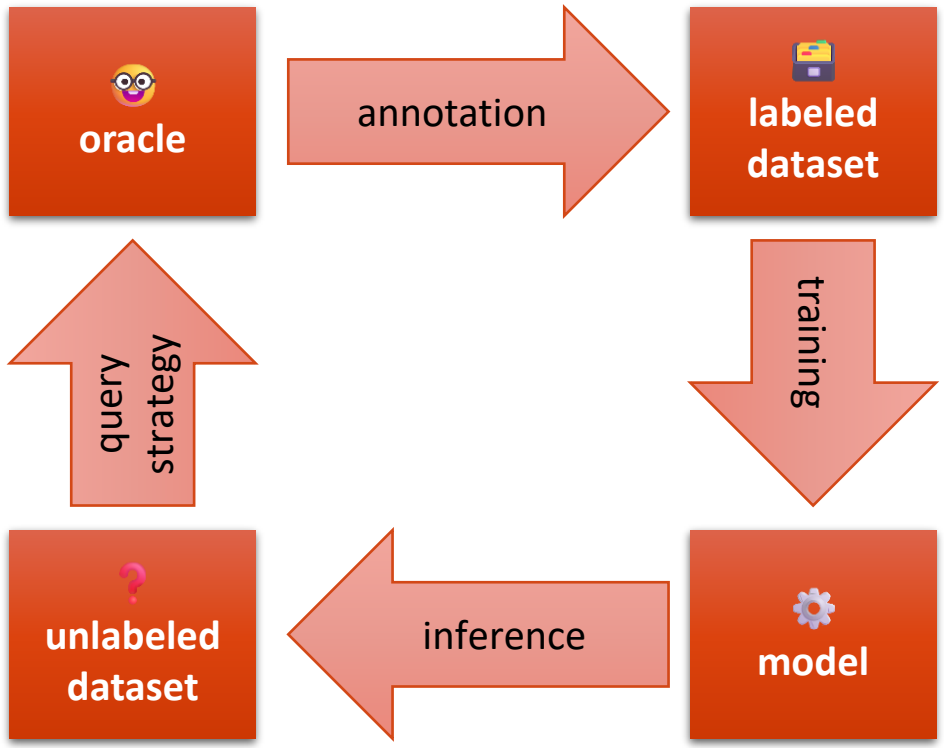
- Deep learning models require vast datasets to generalize well; but, large datasets entail significant manual labor for human annotation.



◁ **Seafood Frustration**
Manual seafood annotation is costly because it requires highly specialized marine biologists to review and label vast datasets. This makes fully supervised training of computer vision models a massive logistical and financial burden.

Solution: Active Learning

- In **active learning**, algorithms automatically identify and select the most ambiguous data points for human review, which reduces overall labeling volume.
- In this paper, the query strategy will be **uncertainty sampling**, which is hypothesized to outperform random sampling.



◁ **General framework for active learning**
The process begins by training the model on a small subset of labeled data. The model then scans the unlabeled dataset for the most informative points based on a query strategy. A domain expert, or oracle, manually labels these selected data points before adding them to the training set. The system then retrains the model iteratively to improve performance. This cycle continues until the model reaches satisfactory accuracy.

Methodology

Dataset and Preprocessing

- **"A Large Scale Fish Dataset"** is one of the most popular and publicly available image classification and segmentation datasets on Kaggle. It has 9 classes, each with 1000 images.

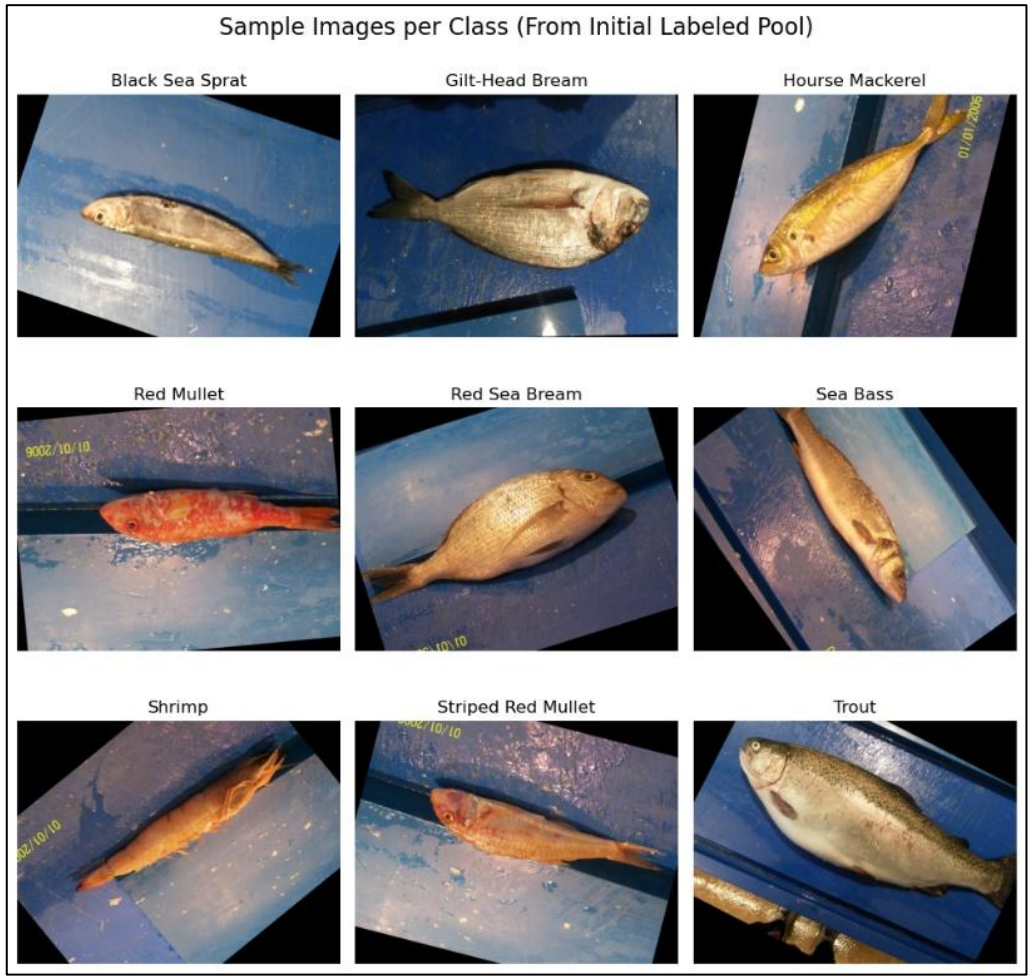
Test / Valid / Train Splits

test set: 200 imgs

valid. set: 100 imgs

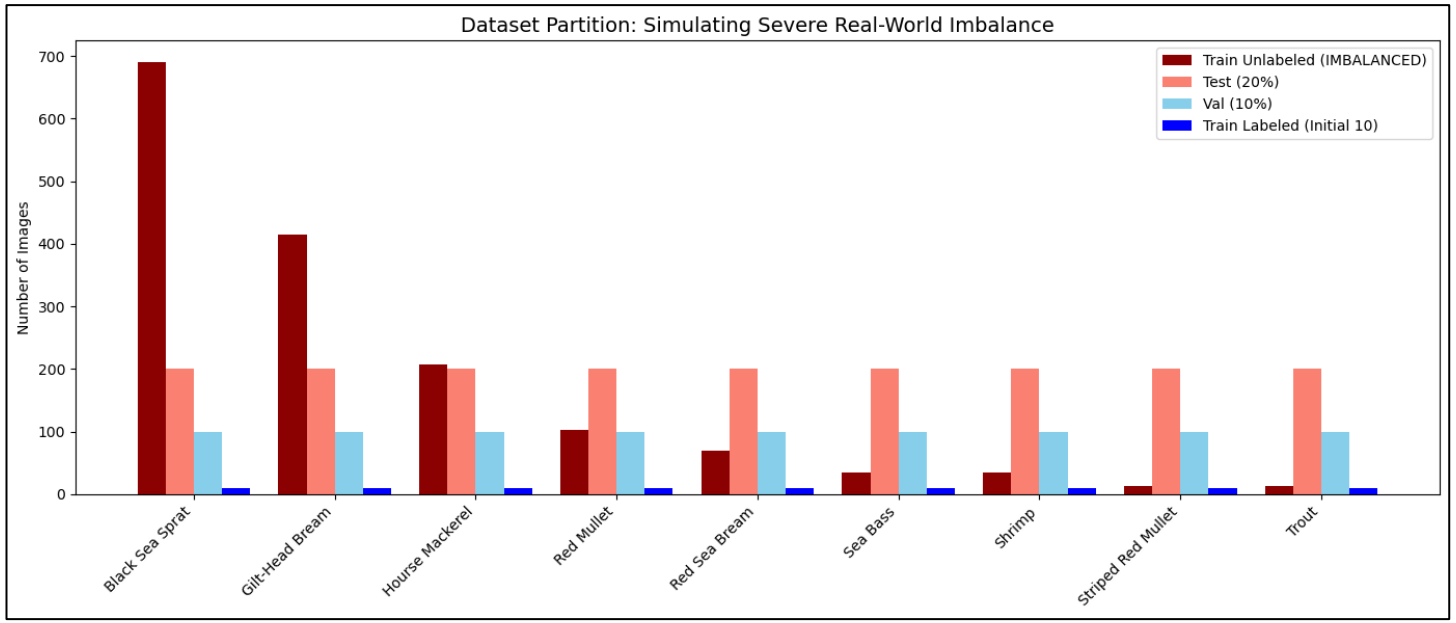
initial labeled train set: 10 imgs

initial unlabeled train set: 690 imgs



◁ **Sample images**
These are just sample frames from each of the classes of "A Large Scale Fish Dataset".

▽ **Simulated imbalanced dataset**
The throttling of the images per class in the unlabeled pool is supposed to simulate a long-tailed distribution. In reality, fish data will vary depending on the species where some are common and some are not. This way, the model will be forced to learn from a heavily imbalanced dataset.



Methodology

Model Architecture

- Using **ResNet-18** for classification within an Active Learning loop is highly strategic because it perfectly balances computational speed, feature extraction quality, and sample efficiency.

Active Learning Strategy

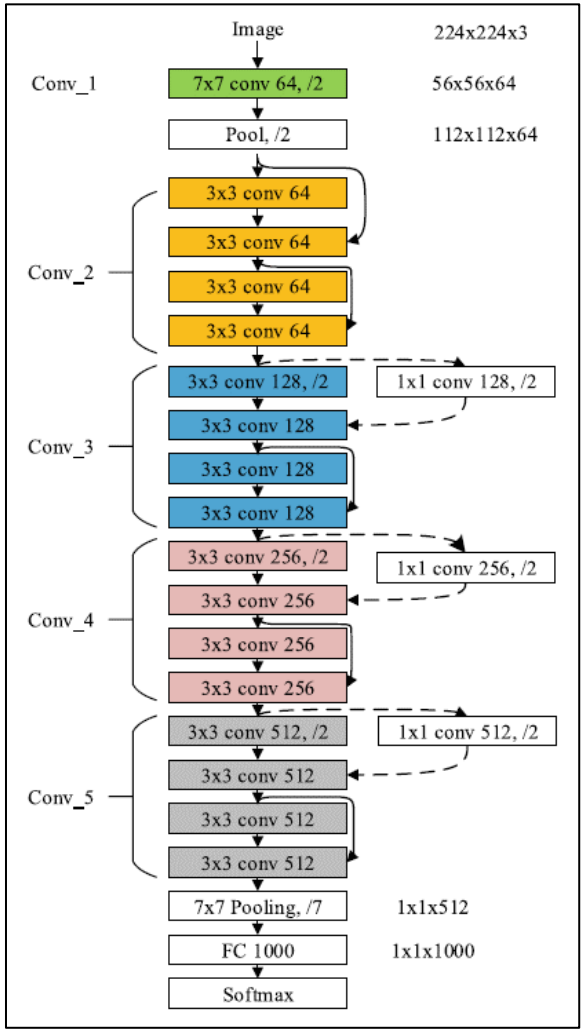
- To iteratively improve the model, a strict query budget of 10 samples per iteration is set across 10 total iterations.
- The training set is ultimately expanded by 100 chosen images.

Experimental Setup

- To rigorously validate the performance of uncertainty sampling, **Random Sampling** is utilized as the baseline control group.
- The network was trained using the **Adam**, **Cross-Entropy loss**, and **StepLR**. Early stopping is also employed.
- To account for noise and ensure the statistical validity of findings, experiments were executed across 3 random seeds: 5, 42, and 101.

ResNet-18

- ResNet-18 is a lightweight, 18-layer convolutional neural network (CNN) primarily used for image classification.
- It features "skip connections" that allow gradients to flow easily through the network.
- Skip connections prevent the vanishing gradient problem and enables efficient training of deeper models.



△ **ReNet-18 Architecture**
ResNet-18 is constructed primarily using Basic Residual Blocks. Source: https://www.researchgate.net/figure/Fig-2-An-illustration-of-the-architecture-of-ResNet-18-model_fig1_383708698

Results

Supervised Learning Baseline

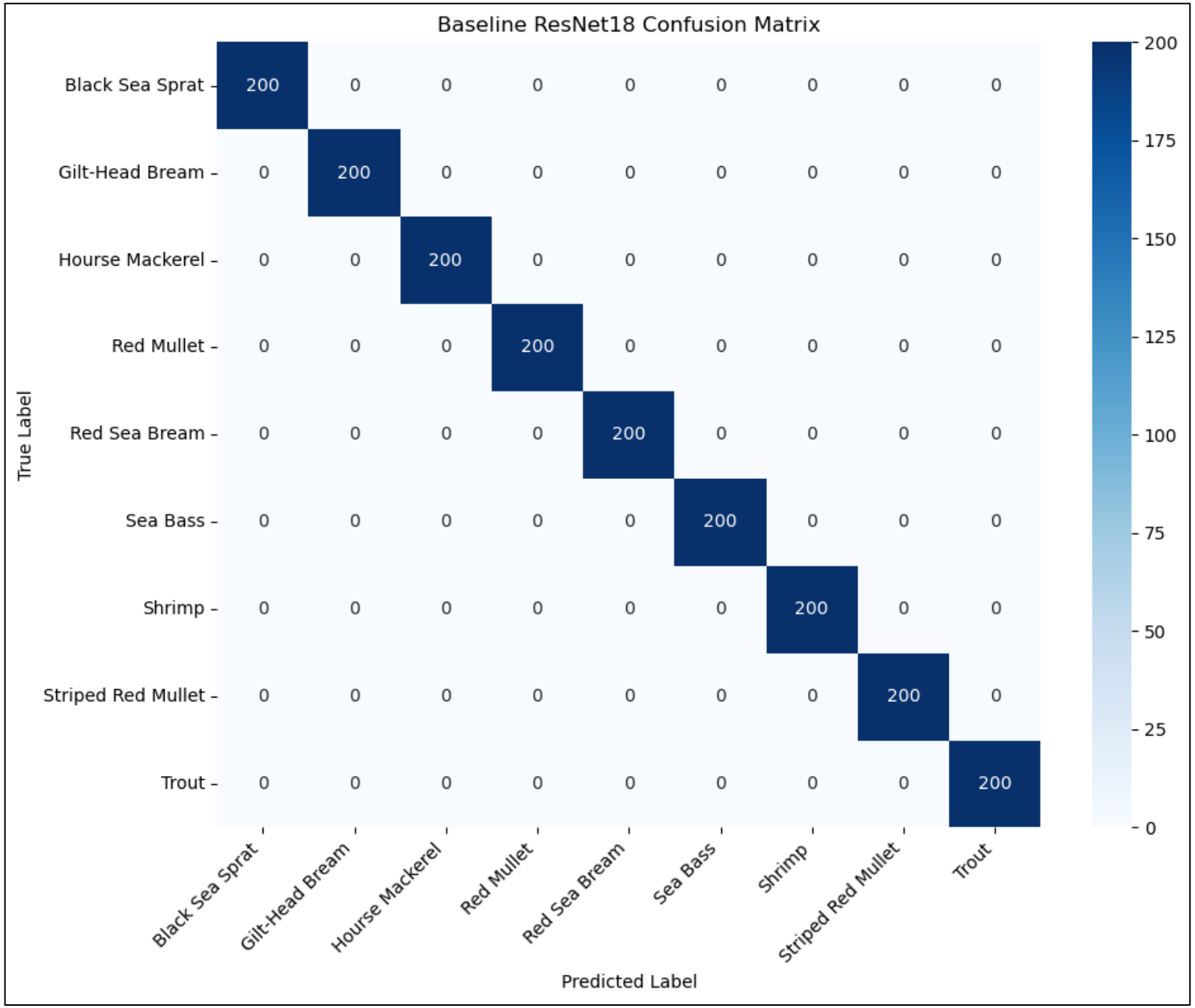
- To establish a theoretical upper bound for model performance, a ResNet-18 model was trained on the entire 70% training pool (comprising $700 \times 9 = 6300$ images).
- The model achieved an exemplary test accuracy of 1.0000 and a test Macro F1 score of 1.0000 as well.

Test Accuracy: 1.0000

Test Macro Precision: 1.0000

Test Macro Recall: 1.0000

Test Macro F1-Score: 1.0000

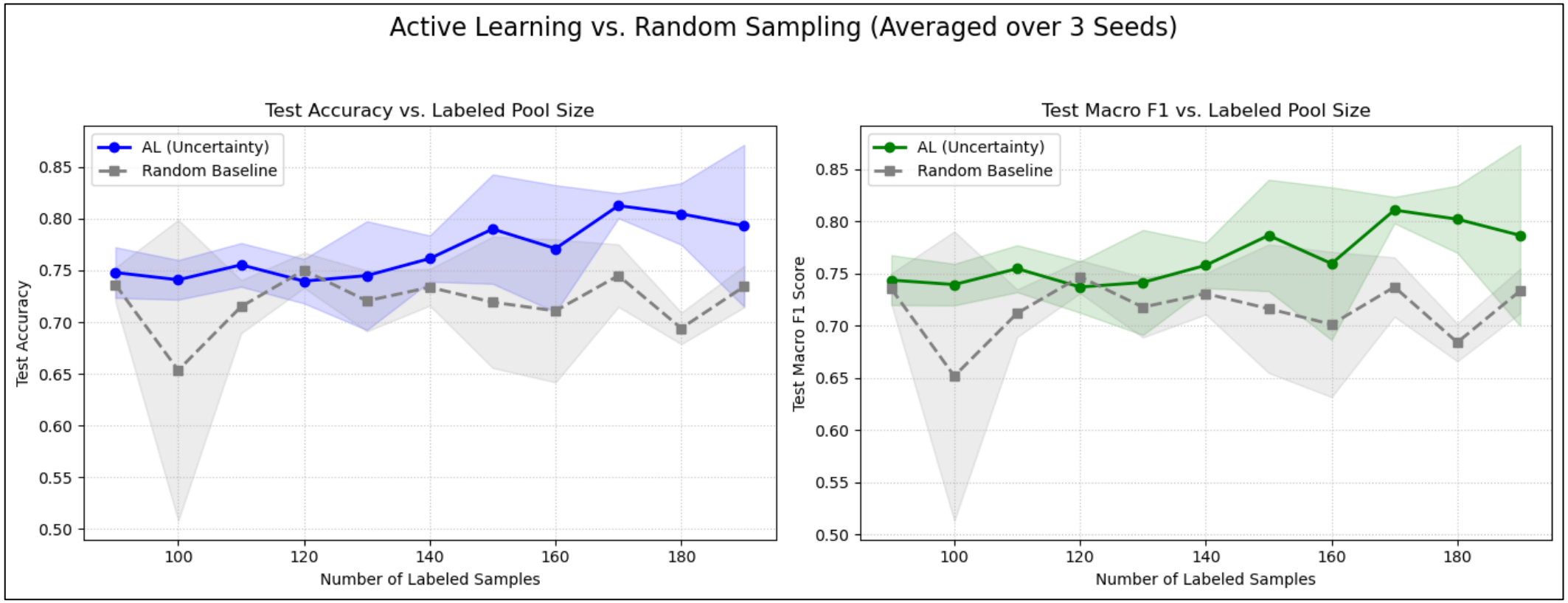


▷ **Confusion matrix**
Baseline ResNet-18 confusion matrix for the supervised learning

Results

Active Learning vs. Random Sampling

- Both strategies exhibited a comparable performance as the labeled pool grew from the initial 90 samples to 190 samples.
- Active Learning demonstrated an upward learning trajectory.
- By Iteration 8, Active Learning achieved a mean Test Macro F1 score of 0.8107 (± 0.0127), whereas the Random Baseline only reached 0.7371 (± 0.0284).

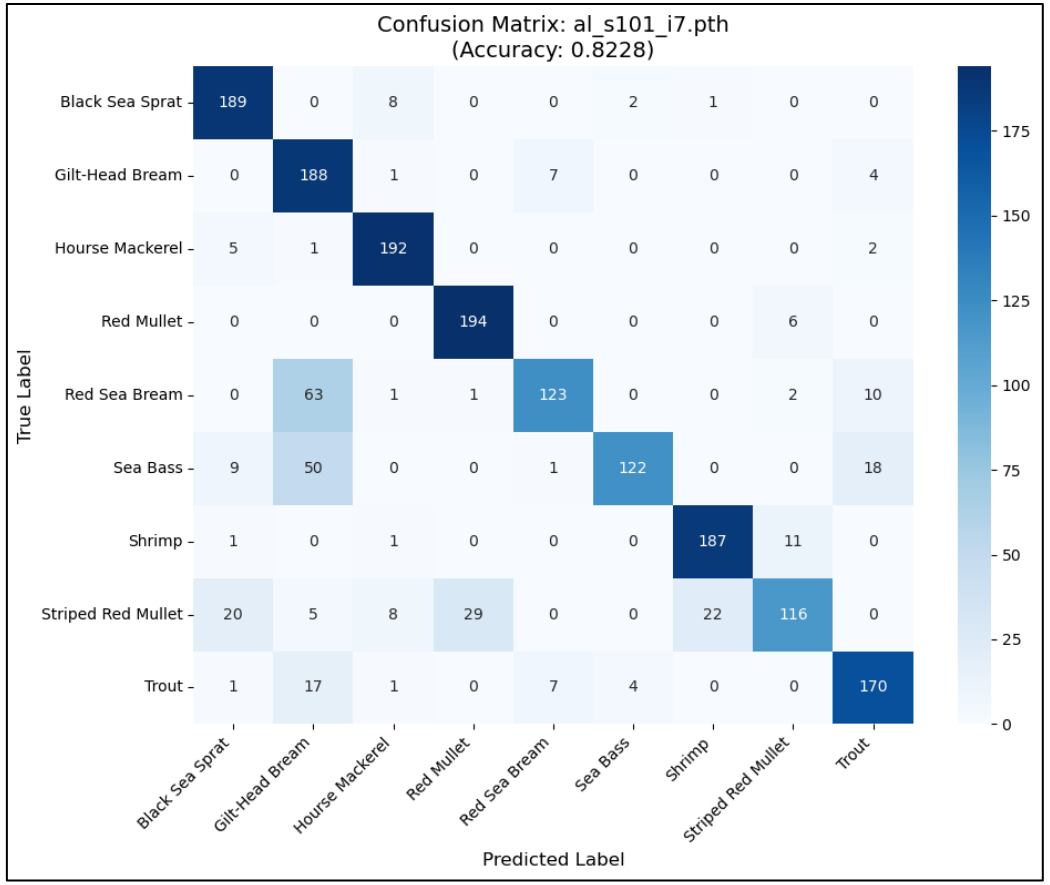


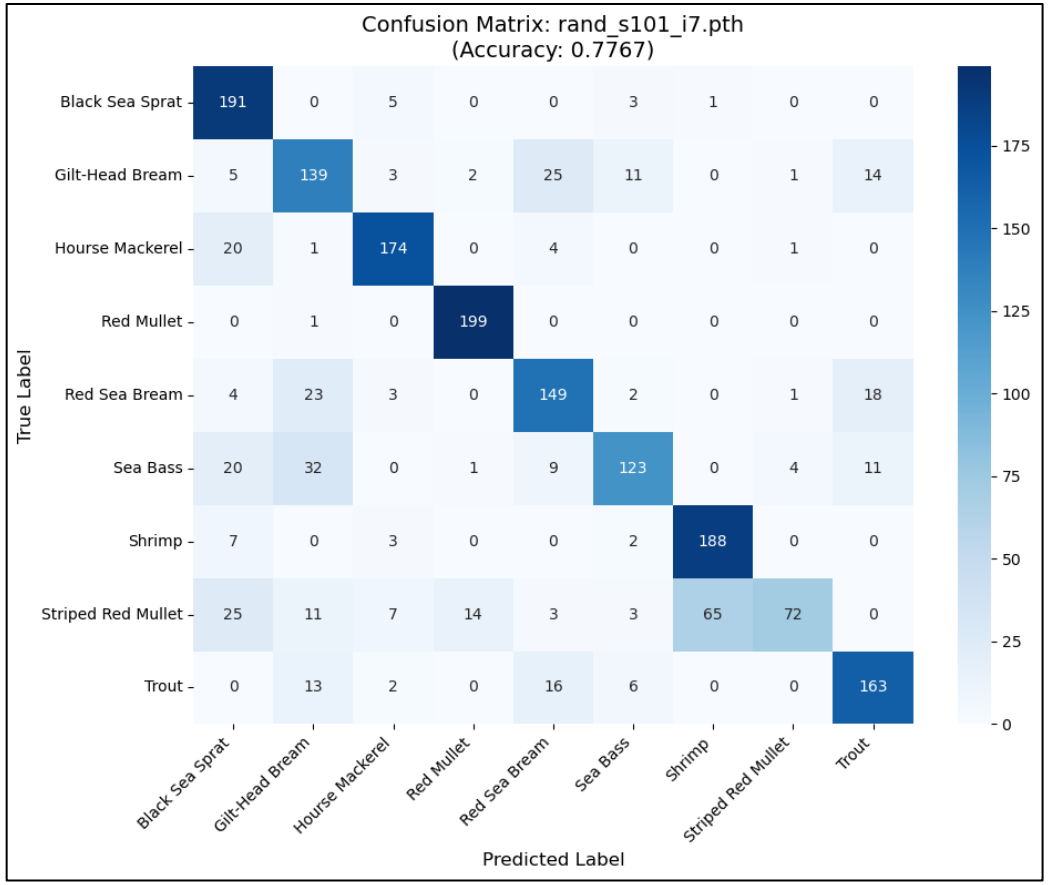
◁ **Learning curves**
As iterations progress, the accuracy and F1-score of Active Learning suggests an upward trend.

Results

Active Learning vs. Random Sampling

- The model developed by Active Learning is confused mostly with the Red Sea Bream being erroneously classified as a Gilt-Head Bream. Meanwhile, Random Sampling appears to be mostly incorrect in classifying Striped Red Mullet as Shrimp.





Results

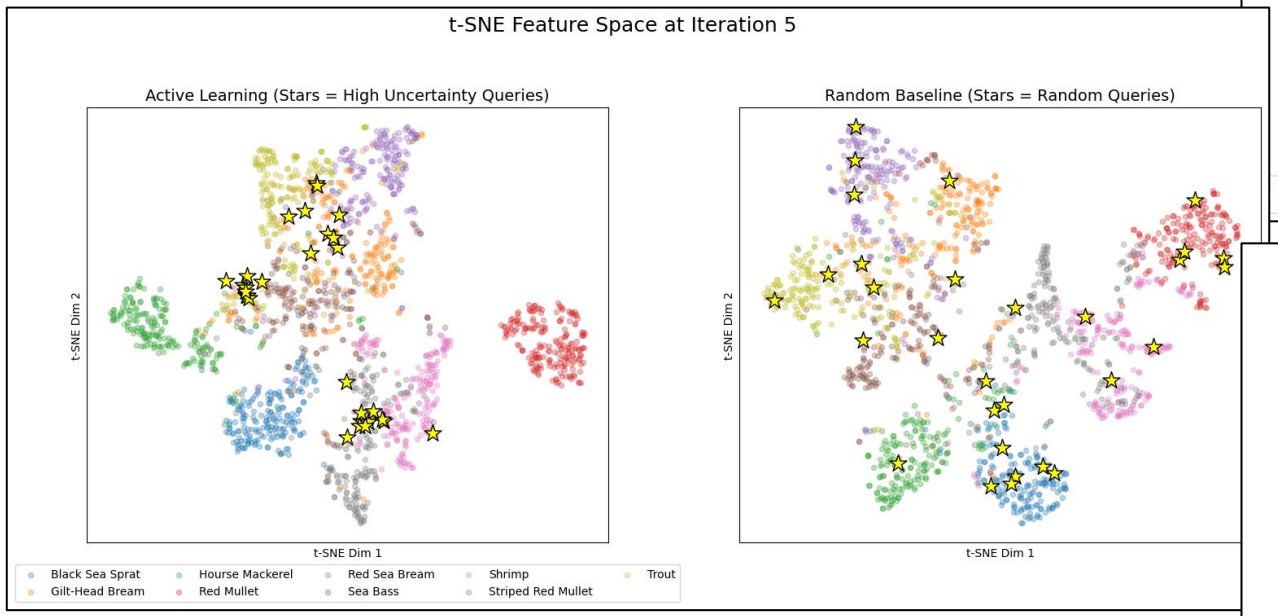
Active Learning vs. Random Sampling

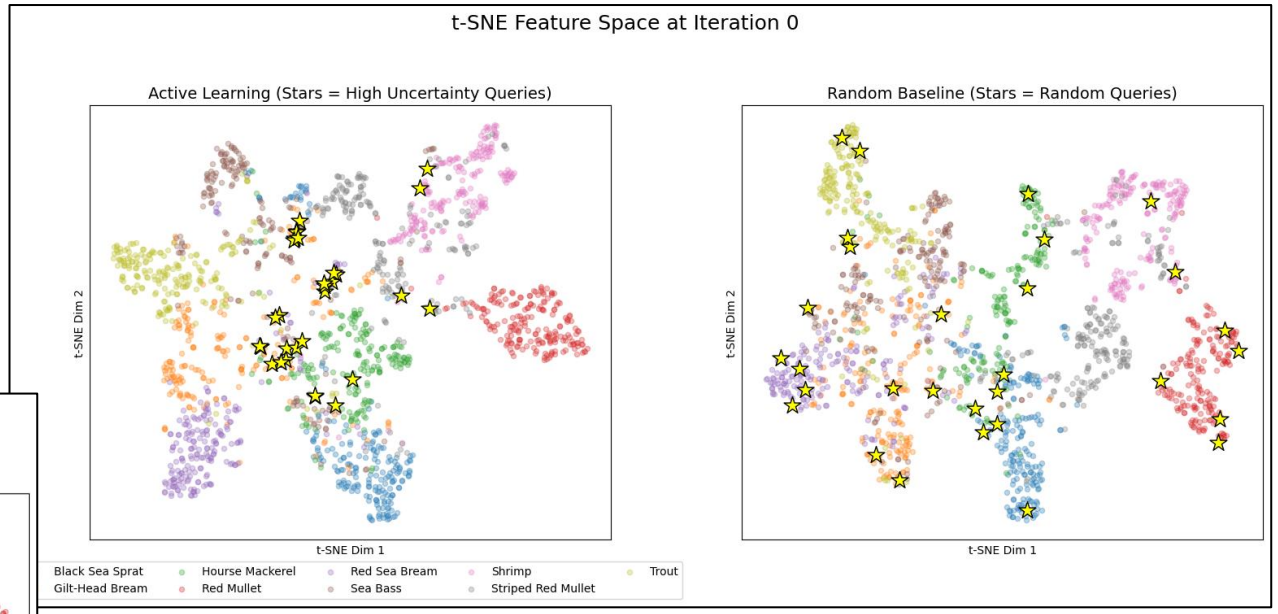
		Active Learning				Random Sampling			
Budget	Iteration	Val_Acc	Val_F1	Test_Acc	Test_F1	Val_Acc	Val_F1	Test_Acc	Test_F1
1.43%	0	0.7448	0.7385	0.7480	0.7437	0.7400	0.7371	0.7363	0.7353
1.59%	1	0.7363	0.7344	0.7409	0.7394	0.6581	0.6554	0.6535	0.6517
1.75%	2	0.7611	0.7584	0.7554	0.7546	0.7293	0.7252	0.7150	0.7119
1.90%	3	0.7496	0.7478	0.7396	0.7370	0.7593	0.7546	0.7502	0.7469
2.06%	4	0.7581	0.7515	0.7450	0.7414	0.7333	0.7289	0.7206	0.7177
2.22%	5	0.7685	0.7650	0.7615	0.7579	0.7552	0.7514	0.7337	0.7307
2.38%	6	0.7859	0.7833	0.7900	0.7865	0.7278	0.7225	0.7193	0.7163
2.54%	7	0.7781	0.7687	0.7711	0.7595	0.7233	0.7110	0.7109	0.7012
2.70%	8	0.8281	0.8261	0.8126	0.8107	0.7496	0.7384	0.7450	0.7371
2.86%	9	0.8137	0.8093	0.8046	0.8020	0.7026	0.6933	0.6941	0.6840
3.02%	10	0.7989	0.7918	0.7931	0.7865	0.7574	0.7563	0.7344	0.7334

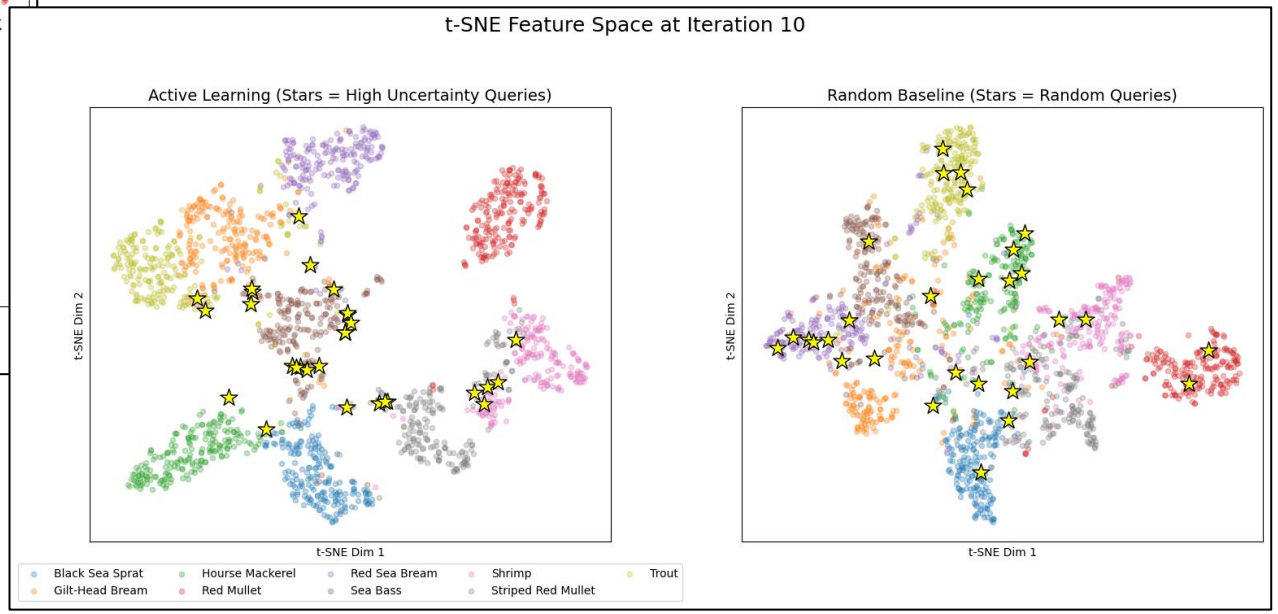
Results

Unsupervised Feature Analysis

- By explicitly querying the most ambiguous or low-confidence images, the Active Learning model appears to disentangle the feature space and refine its decision boundaries with far greater sample efficiency.







Wrapping Up

Limitations

- Query batch size was restricted to 10 samples per iteration.
- Least Confidence only considers the highest-class probability and ignores the rest of the probability distribution.
- Synthetic class imbalance of the training pool.

Future Work

- Explore more sophisticated acquisition functions.
- Incorporate diversity-aware algorithms.
- Explore other models besides ResNet-18.

Summary and Conclusion

- This project demonstrates that Active Learning is a strategy for mitigating the expensive and subjective process of manual data annotation in automated seafood classification.

Problem and Approach

- The project investigated using an **Active Learning framework** with a **ResNet-18** convolutional neural network to **reduce the significant manual labeling costs** associated with heavily imbalanced marine computer vision datasets.

Methodology

- **Uncertainty sampling** was used via the Least Confidence metric to systematically query and **select the most ambiguous unlabeled images** for human annotation, which appears to implicitly helped balance minority classes.

Key Results

- The **Active Learning model outperformed the random sampling baseline**, achieving a Test Macro F1 score of 0.8107 using only 2.70% of the available data, which translates to an approximate 97.3% reduction in required labeling efforts.